

ASSESSING THE PERFORMANCE OF A GREEN LIQUID FUEL HYPERGOLIC WITH HYDROGEN PEROXIDE IN A 50 N BIPROPELLANT THRUSTER

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The last decades have witnessed an increasing interest in the development of green propellants for propulsion systems. Particularly in the low thrust domain, where a feasible alternative to the storable liquid bipropellant system hydrazine – nitrogen tetroxide is yet to be fully demonstrated, the application of green hypergolic propellants such as ethanol and H₂O₂ 90 wt% appears to be a promising one. This work presents the design and firing test of a 50 N bipropellant thruster based on the H₂O₂ 90 wt% – monoethanolamine-ethanol hypergolic pair catalyzed by Cu²⁺, previously developed by our research group. The 5 s firing test, conducted under a swirl injection scheme, yielded average values of 50.38 N, 147 s, and 1210 m/s for thrust, I_{sp} , and c^ , respectively, with the target value of 5 bar for the chamber pressure being reached after 2 s of operation. Alongside with a c^* efficiency of 84%, these results suggest that the assessed green hypergolic bipropellant pair may be a promising alternative for applications in the low-thrust range.*

KEY WORDS: hydrogen peroxide, liquid propulsion, hypergolic, green propellants, monoethanolamine, ethanol

1. INTRODUCTION

Space industry has experienced a remarkable boom in the last few years, motivated to a great extent by the ever-growing dependence of many economic sectors on satellites, not to mention

NOMENCLATURE			
MEBF	monoethanolamine-ethanol based fuel	$\dot{m}$	mass flow rate
		ρ	density
c^*	characteristic velocity		
I_{sp}	specific impulse	Subscripts	
L^*	characteristic length	ch	chamber
p	pressure	ext	exit
F	theoretical thrust	oxi	oxidizer
$R_{O/F}$	oxidizer/fuel ratio	f	fuel
ε	nozzle area ratio	tot	total

the widespread need for weather forecasting, communications, and reliable positioning systems. Military and research satellites are important segments as well. Even space tourism has arisen as a promising field in the next decades (Kang et al., 2016; Pearson, 2004; Sutton and Biblarz, 2016). Of course all of these applications demand dependable launching systems and accurate orbit control technologies, capable of delivering and positioning a diversity of payloads, from large-scale geosynchronous satellites to micro satellites and Cubesats. In an aggressively competitive market the cost-effectiveness and the environmental adequacy play major roles in defining the propulsive system to meet each payload's needs. Environmental friendliness, particularly, has raised serious concerns (Anflo and Möllerberg, 2009; Brinck, 2014) due to the increasing tendency in the number of launches combined with the lasting dependency on heritage technologies.

Regarding upper-stage engines, attitude adjustment systems in satellites, reaction control devices on spacecrafts and small propulsion systems, storable liquid bipropellants meet performance requirements hardly attainable by other propulsion alternatives. The state of the art storable bipropellant system comprises hydrazine and its derivatives as fuels and nitrogen tetroxide (NTO) as oxidizer. The key feature for its extensive application in small-scale engines, in the thrust range from 5 to 4000 N (Sutton and Biblarz, 2016) relies upon its hypergolicity, i.e., the low-temperature self-ignition provided by the contact between fuel and oxidizer (Pichon et al., 2005). This means that no ignition devices are required, favoring the system mass ratio, and that a multiple start capacity is achievable.

In spite of their excellent performance, good thermal stability, low shock sensitivity, and data availability, hydrazines–NTO systems represent a major concern for its effects on human health and on the environment. They exhibit an unacceptable level of toxicity, high vapor pressure, and volatility. Moreover, hydrazine and its derivatives are carcinogenic (Brinck, 2014; Pichon et al., 2005).

Environmentally feasible alternatives to the storable hydrazines–NTO systems are yet to be fully demonstrated. This lack of consistently viable options results from the simultaneous requirements that should be met by the candidate substitutes, for instance: non-toxicity, non-carcinogenicity, safe handling, long-term storability, low vapor pressure, hypergolicity, and acceptable cost. Some investigations have considered solely the replacement of the fuel, keeping NTO as the oxidizer (Pichon et al., 2005), while others (Kang et al., 2016; Kang and Kwon, 2017, 2018) have considered the problem from a wider and more appropriate point of view, studying new alternatives for both the fuel and oxidizer. Although these studies have examined

some exotic chemical species such as tetrahydrofuran and toluene (Filley et al., 2004; Fowles et al., 2013), they have suggested rocket-grade hydrogen peroxide (RGHP) as a feasible green oxidizer, capable of hypergolic ignition with the developed fuel formulations.

RGHP has proved itself as an effective green oxidizer in combination with proper fuels like Block 0, developed by the US Naval Air Warfare Center. It comprises a colloidal suspension of manganese oxide in methanol, making it a catalytic hypergolic fuel (Guseinov et al., 2018; Lauck et al., 2019). Despite its remarkable hypergolic performance, the solid catalyst amount is expressive (10 to 30 wt%). Melof and Grubelich (2001) have assessed a number of fuels as hypergolic pairs to H_2O_2 90 wt%, the best result being achieved with ethanolamine catalytically promoted with CuCl_2 .

In a previous study, our research group has presented the successful optimization of a fuel made of monoethanolamine and ethanol, catalyzed by copper nitrate trihydrate (MEBF) (Maschio et al., 2018), hypergolic with H_2O_2 90 wt%. Its composition has been defined through a DoE study aiming at minimizing the ignition delay. The ignition mechanism has been qualitatively described as well, emphasizing the roles played by each of the fuel's components. Therefore, this work intends to design and test a 50 N bipropellant engine based on the mentioned hypergolic pair. This is a necessary proof of concept of its feasibility as an upper-stage engine and/or an attitude/reaction control system.

2. METHODOLOGY

Stabilized H_2O_2 70 wt% (Interox 70-10) provided by Peróxidos do Brasil (Solvay Group) was concentrated up to a 90 wt% through a counterflow drying process employing hot air. The concentrator comprised a Jacob's ladder tube heated in a water bath at 370 K. Hot dry air was introduced in one side of the tube, while H_2O_2 70 wt% was injected in the other side. The H_2O_2 concentration was checked by titration in acidic medium with a standardized potassium permanganate solution. No additional stabilization beyond that provided by the supplier was applied to the concentrated material.

A previous study has investigated the development and optimization of a monoethanolamine, ethanol, and copper nitrate-based rocket fuel, hypergolic with H_2O_2 90 wt% (Maschio et al., 2018). The fuel composition was set according to a DoE approach, aiming at minimizing the ignition delay. The optimized fuel composition was 61.0 wt% of monoethanolamine, 30.1 wt% of ethanol, and 8.9 wt% of copper nitrate trihydrate.

The first propulsive assessment of the optimized hypergolic pair required the design of a thrust chamber with a modest thrust level, capable of providing, simultaneously, the necessary insight into their real performance and acceptable safety conditions. Thus, the following design parameters were defined: 50 N theoretical thrust, 500 kPa chamber pressure, 96 kPa ambient pressure, optimum oxidizer to fuel mass ratio (O/F ratio), nozzle diverging section with 7.5° half-angle, and 1 MPa injection pressure. The calculations were then performed with RPA Standard Edition v.2.3.2 software, based on these initial settings, adopting the additional hypothesis of one-dimensional isentropic flow in equilibrium condition during expansion. The complete output is depicted in Table 1.

Dimensioning the thrust chamber required first an estimate of the characteristic length according to the calculations described by Barrère (1960). This initial estimate yielded a length of nearly 1.5 m. In order to simplify chamber machining, a conical nozzle was selected, with half-angles of the converging and diverging sections of 30° and 7.5° , respectively. Table 2 describes the configuration of the thrust chamber.

TABLE 1: Initial propulsive settings and RPA output for the 50 N bipropellant thruster

Parameter	Value
p_{ch} (kPa)	500.00
p_{ext} (kPa)	96.00
F (N)	50.00
Optimum $R_{O/F}$	3.72
ε	1.55
c^* (m/s)	1518.02
I_{sp} (s)	170.10
$\dot{m}_{tot}$ (g/s)	30.00
$\dot{m}_{oxi}$ (g/s)	23.68
$\dot{m}_f$ (g/s)	6.32

TABLE 2: Thruster dimensions

Parameter	Value
Nozzle type	15° conical nozzle
Nozzle throat diameter	10.83 mm
Nozzle expansion area ratio	1.55
Chamber length without converging section	68.27 mm
Chamber length with converging section	113.76 mm
Converging half-angle	30°
Chamber diameter	43.32 mm

A dual swirl injector type was chosen, dimensioned according to the procedures described by Savonov (2011). In short, the fuel injector comprised a vortex chamber and three tangential entrance orifices (0.42×0.42 mm), while the oxidizer injector was made of a vortex chamber and six tangential entrances (0.60×0.72 mm). The thrust balance consisted of an external stainless-steel frame, flexible spring steel sheets 0.2 mm thick and a load cell, coupled to a signal amplifier and mounted on the injection subassembly.

The feeding system included a commercial compressed nitrogen gas cylinder to pressurize the propellants, storage tanks for the oxidizer and the fuel, connected to stainless-steel feeding tubes, as well as other required accessories depicted in Fig. 1. The feeding lines were pressurized in order to attain a 1 MPa manometric figure at the injectors, with measurement taken in the injection chamber, by means of a Novus NP400 pressure transducer. Chamber pressure was taken by another Novus NP400 pressure transducer, placed 20 mm away from the injector exit. Flow rates were provided by Flomec OM006S511 flow meters. Injection valves (Swagelok SS-42GS4-SC11-31CD) operation was conducted with a microprocessed digital timer COEL TT34, remotely triggered. Data acquisition was performed through a Novus FieldLogger board. Figure 1 depicts schematically the test stand.

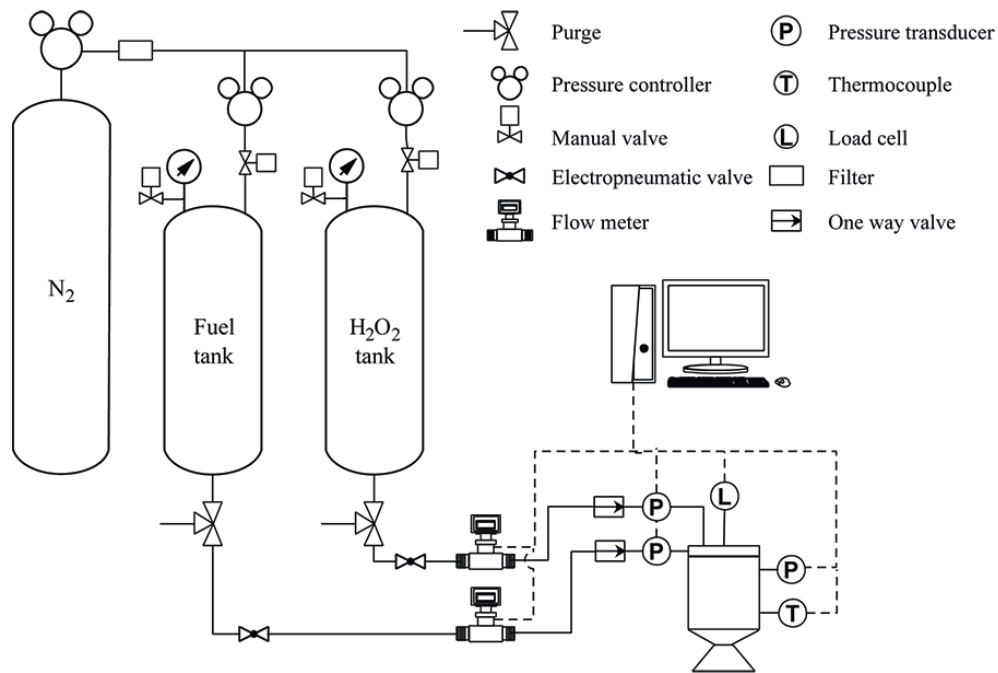


FIG. 1: Schematic representation of the test stand.

3. RESULTS

In liquid propulsion, the injection system plays a vital role in the atomization and in the combustion processes of the propellants. As long as combustion efficiency is extremely dependent on the propellants atomization and intimate mixing, a proper thruster performance depends extensively on the injectors (Sakaki et al., 2017). Therefore, it was necessary to evaluate beforehand the injector's behavior under the same pressure gap envisaged for the thruster, i.e., 5 bar. Such an assessment was carried out by means of high-speed imaging.

The pictures were acquired with a FASTEC TS3100SC4 high speed camera at a 2000 fps rate. It was unfeasible to acquire the images with the propellants themselves, due to their hypergolicity and the resultant flames. Therefore, in order to get a reliable description of the flow pattern induced by the injector, water was an obvious choice as a surrogate for hydrogen peroxide, for they are known to behave in an analogous fashion from a rheological standpoint. The results are shown in Fig. 2. The first picture (upper left) shows an approximately steady flow pattern consisting only of fuel. The second picture (upper center) illustrates the opening of the oxidizer valve, with water injected through the internal orifice, flowing in the center of the nearly ellipsoidal sheet defined by the fuel. The third one (upper right) exposes the beginning of the interaction between the fuel and oxidizer (water as a surrogate) flows, characterized by the disintegration of the fuel flow. Despite the fact that specialized techniques for the drop size evaluation were not used, the subsequent pictures (bottom left to right) clarify the formation of a homogeneous conical spray, from a macroscopic point of view, as a result of the concentric injection scheme of the propellants. Therefore, an adequate level of atomization was attained by the injection system.

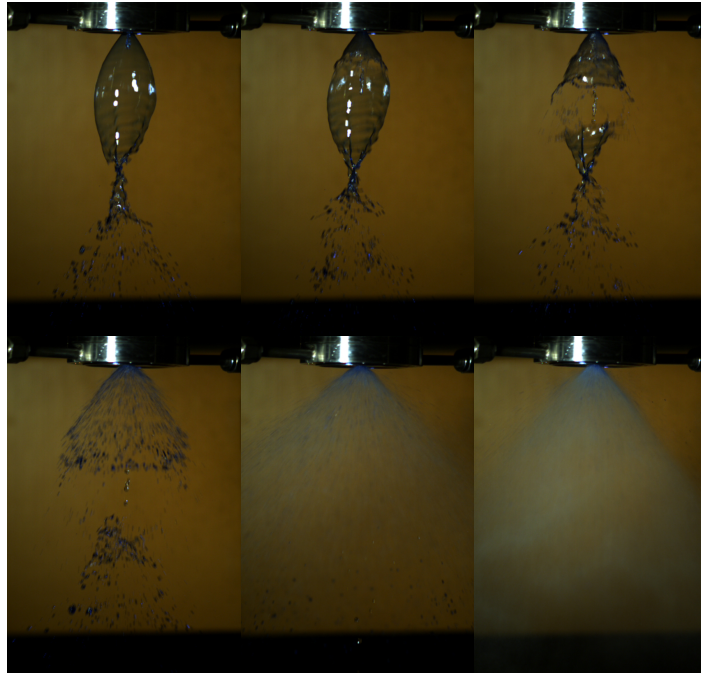


FIG. 2: Injector performance assessment

The next step was the thruster assessment. As a safety precaution, it was assembled in vertical position in order to prevent a catastrophic failure due to a possible accumulation of propellants inside the chamber in the event of a malfunction of the thruster. Figure 3 shows the thruster in steady state operation during a 5 s firing.

Regarding the injection pressure initially set (1 MPa), the measured propellants mass flow rates were roughly 9.8 and 25.8 g/s for the fuel and oxidizer, respectively. It was found that the fuel mass flow rate was different from the calculated one (6.32 g/s). This could stem from the precision level attainable by the machining process of the injector. Taking into account the low thrust level, the injection orifices are tiny. As a result, little deviations from the specified dimensions are prone to affect the flow rates. Obviously, this condition could have been circumvented through an injection pressure reduction. Nevertheless, considering the ranges of the chamber and injection pressures as well as the consequent risk of propellant counter flow, such an alternative was unfeasible in this study. The measured pressure and thrust-time traces are shown in Fig. 4. The characteristic velocity, c^* , and the specific impulse, I_{sp} , stem from these time traces and are shown in Fig. 5.

In spite of the satisfactory agreement between the theoretical and experimental values for chamber pressure and thrust, c^* and I_{sp} exhibited lower values as compared to the initial calculation settings (Table 1). This discrepancy cannot be linked to the combustion and expansion efficiencies listed in Table 1. The augmented fuel mass flow rate can thus be considered an assignable cause for the c^* and I_{sp} discrepancies, for it leads to a lower O/F ratio in comparison to the optimum one, therefore yielding an operational condition different from the one initially set. The fuel mass flow rate of 9.8 g/s measured during the firing resulted in an O/F ratio decrease from the 3.72 optimum value to 2.6, lowering, as a consequence, c^* and I_{sp} .

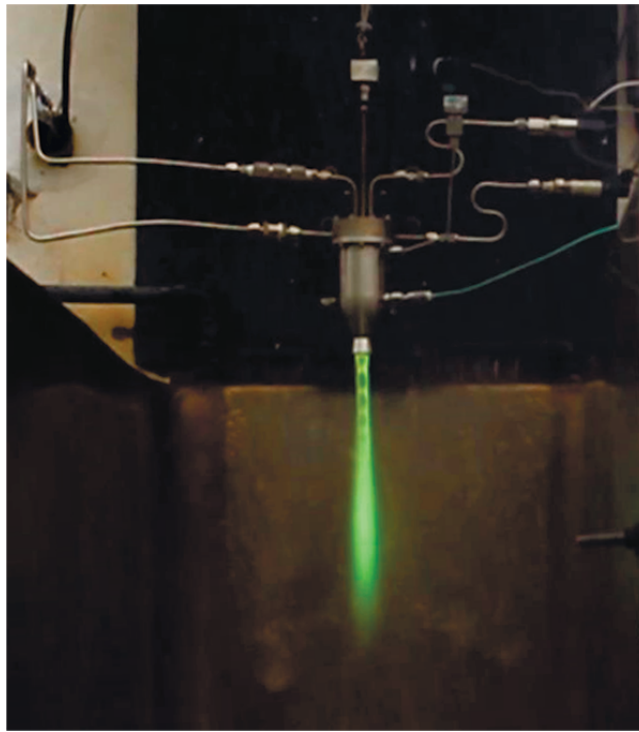


FIG. 3: 50 N bipropellant thruster in steady state operation

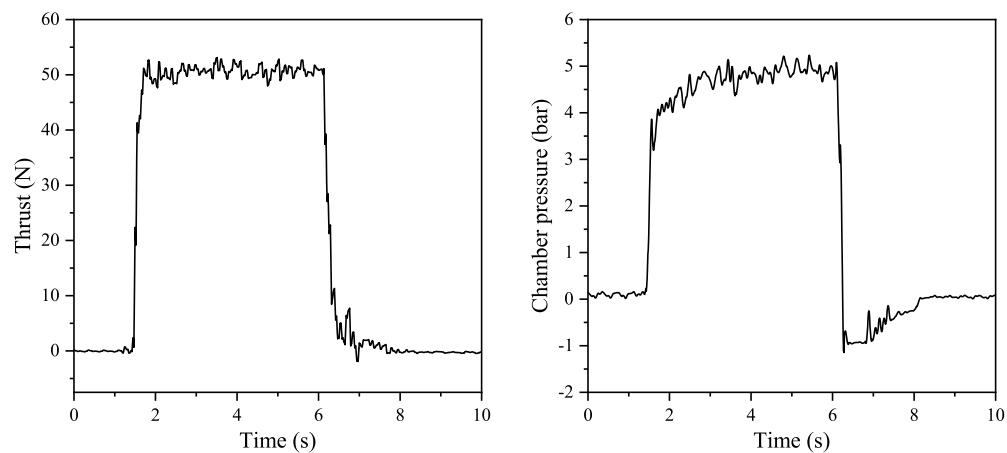


FIG. 4: 50 N bipropellant thruster measured pressure and thrust-time traces

The new O/F ratio setting of 2.6 led to the recalculation of the thruster settings, this time taking as an input the measured total mass flow rate of 35.6 g/s, while the remaining thruster parameters were kept constant. This new estimate was necessary to provide a meaningful basis of comparison between theoretical and experimental values, given the unexpected fuel mass flow

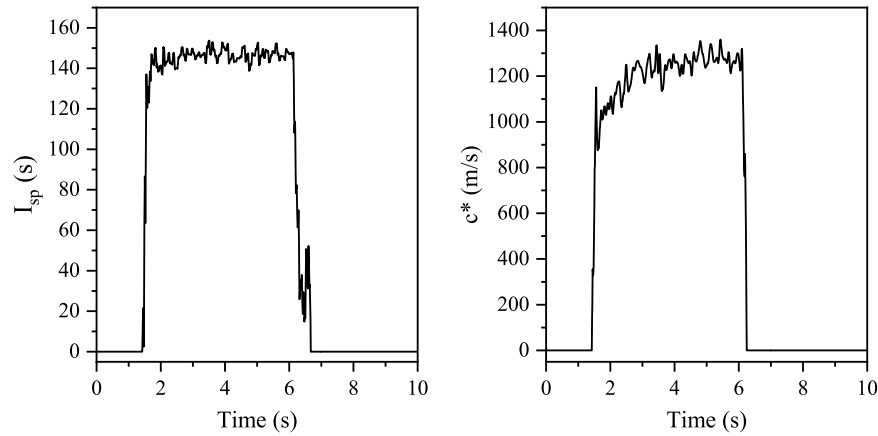


FIG. 5: Characteristic velocity and specific impulse-time traces

rate rise. Thus, theoretical c^* decreased from 1518.02 m/s to 1441.96 m/s and I_{sp} from 170.10 s to 161.79 s.

The thrust and I_{sp} time traces provided the average values of 50.38 N and 147 s, respectively, taking into account a time lapse of 5 s, which corresponds to the total firing time and therefore includes the initial transient period of pressure build-up. Likewise, from the c^* curve it was possible to get the experimental average c^* of 1210 m/s. These figures agreed satisfactorily with the corresponding theoretical estimates for c^* and I_{sp} , as well as with the thrust and chamber pressure initially set for the engine, i.e., 50 N and 5 bar. Subsequently, c^* efficiency amounted to 84%, a quite acceptable value in view of the probably mild fuel-oxidizer mixing effectiveness attained. Considering the 15 ms ignition delay provided by the hypergolic bipropellant pair assessed in this study by means of drop tests, the observed transient period of nearly 2 s to achieve the designed chamber pressure suggests that fuel-oxidizer mixing could be further improved by changing the swirl injector design or even by choosing another injector configuration. Nevertheless, the 84% c^* efficiency is satisfactory, regarding other bipropellant systems recently described (Asakura et al., 2018; Hollingshead et al., 2019).

The next step was to compare the green hypergolic pair considered in this study with some representative bipropellant systems, frequently used in space missions (Sutton and Biblarz, 2016), taking I_{sp} and density- I_{sp} as figures of merit. Table 3 summarizes these data, generated by RPA under the same hypothesis previously mentioned.

This green hypergolic pair offers a quite competitive value for density- I_{sp} , comparable to those attained by some classical pairs like $N_2H_4-N_2O_4$. Unlike these widely employed bipropellant systems, the H_2O_2 90 wt% – MEBF pair does not feature the human health and environmental unfriendliness typical of N_2H_4 , MMH, and N_2O_4 . This is a remarkable advantage given the ever more rigid legislation constraints regarding the environmental effects of space activities (Brinck, 2014).

Other paramount aspects are the storability and simplicity of the green hypergolic pair when compared to the H_2-O_2 system, particularly in applications such as upper stages of commercial space launchers, whose envelope constraints can be met by flexible bipropellant systems, capable of performing accurate orbit insertion maneuvers due to their throttability (Sutton and Biblarz,

TABLE 3: Theoretical I_{sp} and density- I_{sp} comparison for a number of bipropellant systems ($p_{ch} = 500$ kPa and $p_{ext} = 96$ kPa)

Oxidizer	Fuel	Opt. $R_{O/F}$	ρ_{oxi} (kg/m ³) [†]	ρ_f (kg/m ³) [†]	I_{sp} (s)	ρI_{sp} (10 ³ kg.s/m ³)
LOX	Ethanol	1.52	1140.0	789.8	190.6	190.8
LOX	RP-1	2.21	1140.0	807.0	198.0	205.1
H ₂ O ₂ 90 wt%	Ethanol	4.30	1400.0	789.8	176.2	226.4
H ₂ O ₂ 90 wt%	MEBF	3.72	1400.0	1150.0	170.1	229.2
N ₂ O ₄	MMH	1.75	1445.0	875.3	194.2	240.4
LOX	LH2	3.30	1141.2	71.3	271.9	242.6
H ₂ O ₂ 90 wt%	N ₂ H ₄	1.91	1400.0	1008.0	188.8	243.9
N ₂ O ₄	N ₂ H ₄	1.09	1445.0	1008.0	198.4	245.3

[†]Density values are those available at RPA data files, except for H₂O₂ 90 wt% and MEBF, whose densities were experimentally measured.

2016). The combined storability and simplicity may reduce the overall system cost. Finally, the application of the H₂O₂ 90 wt% – MEBF pair in ignition devices might be considered under the prospects of reliability and performance, for instance, in hybrid systems, whose effective ignitability may sometimes pose difficulties (Chiaverini, 2007).

4. CONCLUSIONS

A 50 N bipropellant thruster based on the H₂O₂ 90 wt% – MEBF hypergolic pair was successfully designed and tested in a 5 s firing, under a swirl injection scheme, yielding average values of 50.38 N, 147 s, and 1210 m/s, for thrust, I_{sp} and c^* , respectively, thus exhibiting a performance compatible with the theoretical predictions. The c^* efficiency amounted to 84%, a quite satisfactory outcome considering that the 5 bar chamber pressure set as a target value was attained after 2 s of the thruster operation. A revision of the injector design parameters or even the choice of a different injection approach might further improve c^* efficiency. The hypergolicity, storability, low cost, environmental friendliness, and simplicity of the H₂O₂ 90 wt% – MEBF system makes it a promising alternative for applications in the low-thrust range, such as upper-stage engines, apogee motors, and attitude/reaction control devices for satellites and spacecrafts.

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